

Data Visualization with GGplot2

Lesson objectives

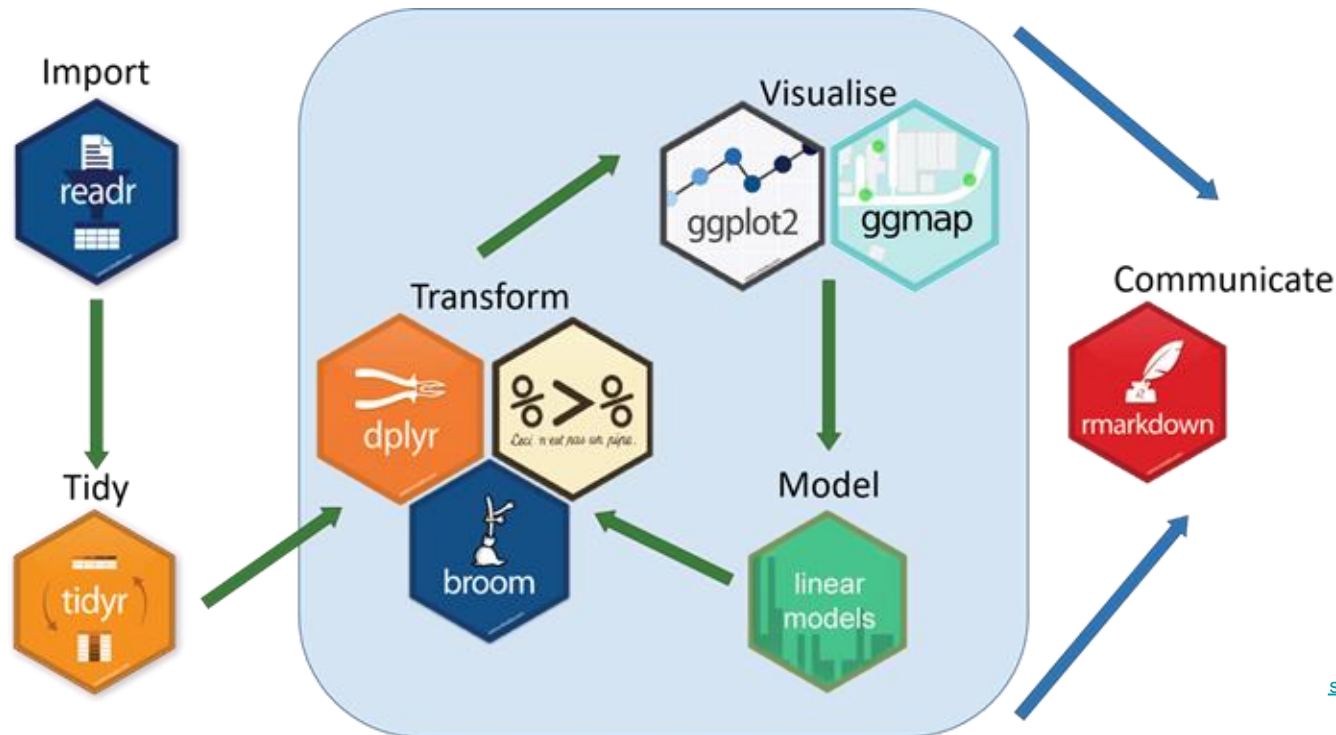
By the end of this session, you'll be able to:

- Locate data visualization within the data science workflow
- Use the grammar of ``ggpplot2`` to create visualizations in R
- Make four different plots using ``ggpplot2`` to visualize an example dataset

Data visualizations

Data visualization is an important part of exploring, understanding and sharing data. Looking at data in a graphical format is a critical part of the data science workflow.

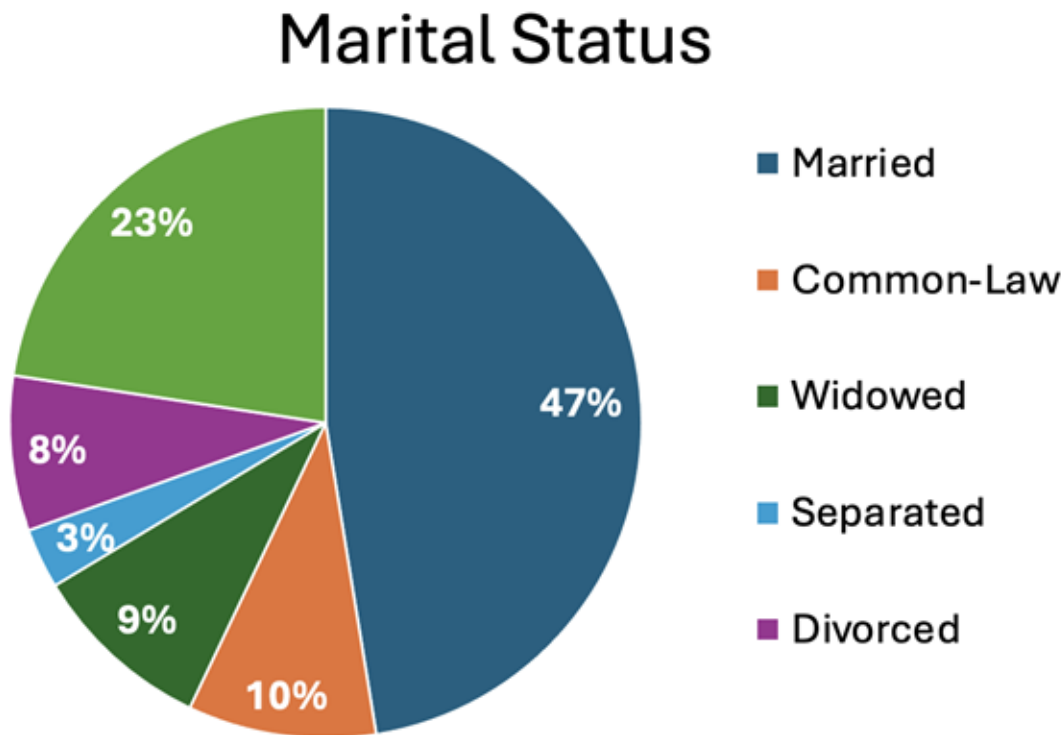
Data visualizations



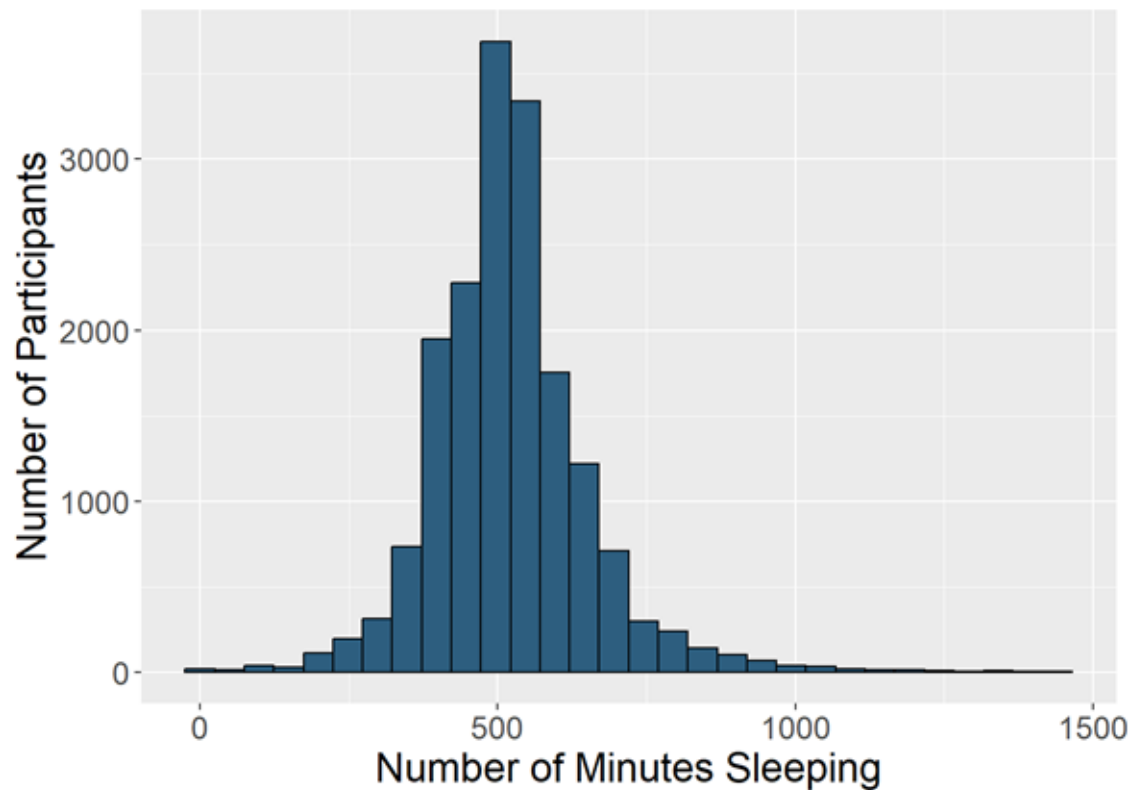
Types of visualizations - Tables

	Mean (minutes)	SD
Time Driving	44.09	65.88
Time Working	167.88	246.12
Time Alone	636.47	475.68

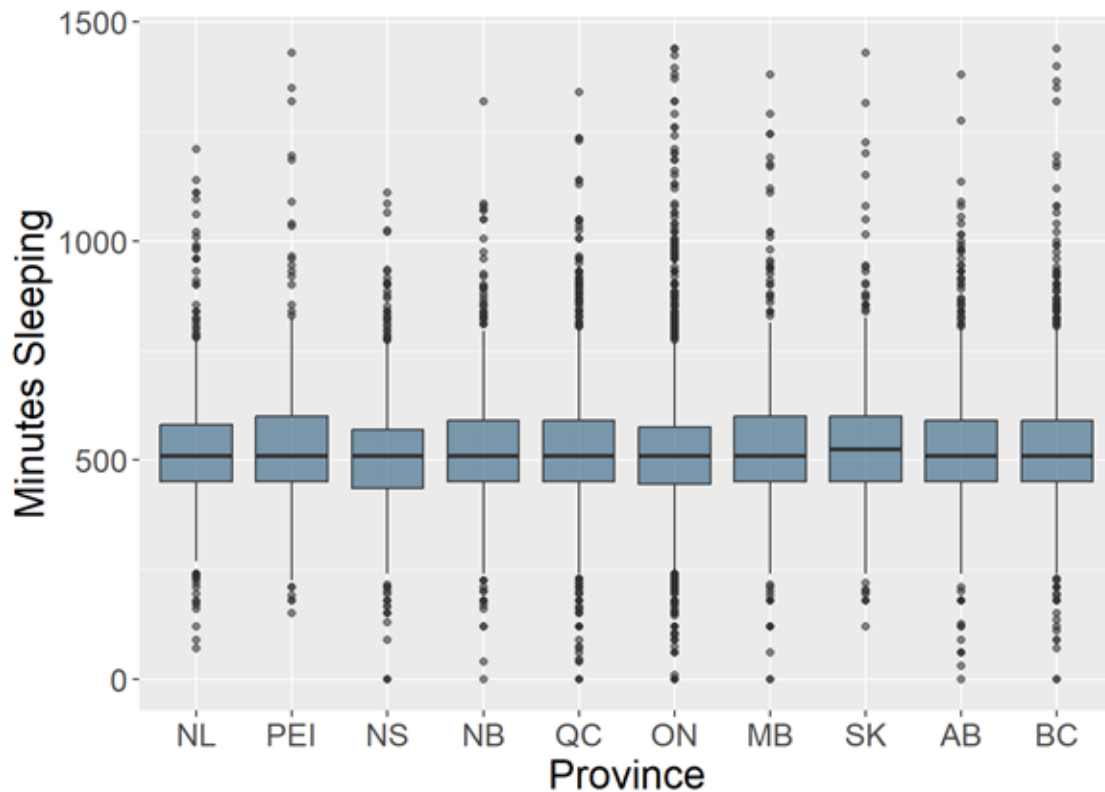
Types of visualizations - Pie charts



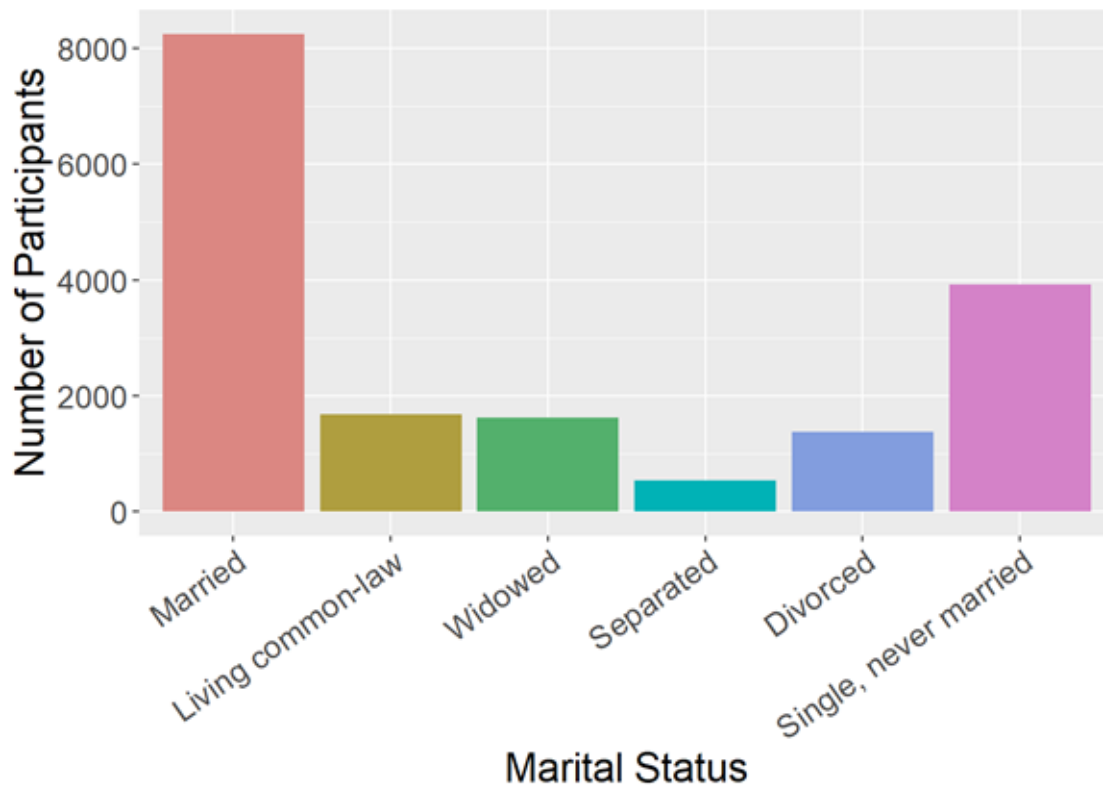
Types of visualizations - Histograms



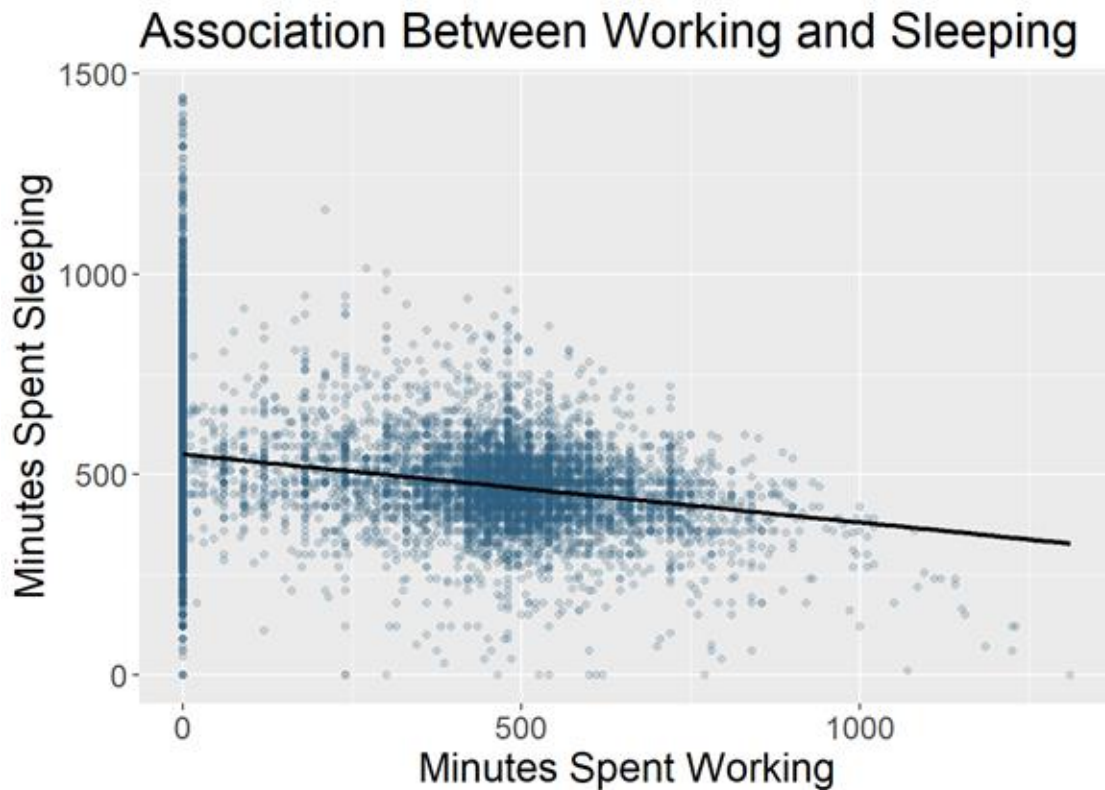
Types of visualizations - Box plots



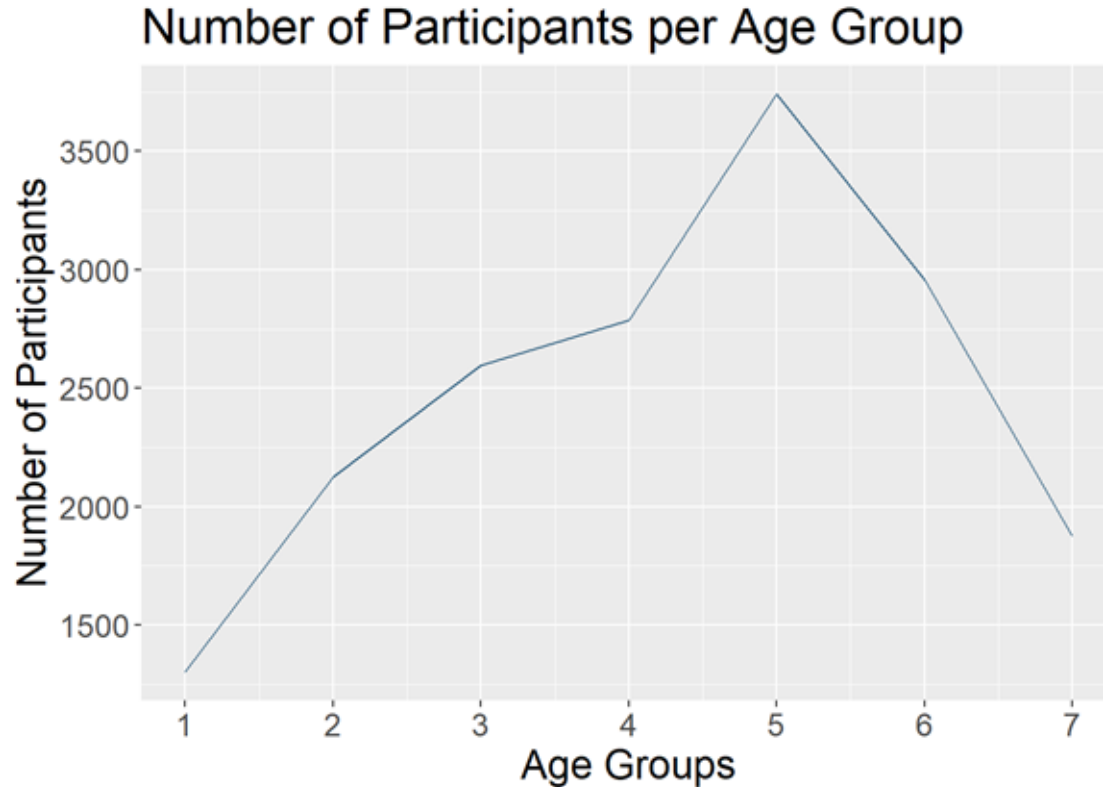
Types of visualizations - Bar charts



Types of visualizations - Scatter plots



Types of visualizations - Line charts



Types of visualizations

And many more!

- Heat Map
- Stacked Bar or Stacked Area Chart
- Violin Plot
- Gantt Chart
- Choropleth Map
- ...

What makes a good visualization?

What makes a good visualization?

- What question are you exploring with your data and how will it inform future analyses?
- What is your data visualization adding to your science communication?
- What take-away message are you conveying with the image?
- What type of variables do you have? Continuous? Categorical? And what type of abstraction (e.g., color, shape) best suits that variable?
- Are you comparing groups?
- Are you showing change over time?
- Are you visualizing a relationship between variables?

Let's jump into R!